

GNU Emacs

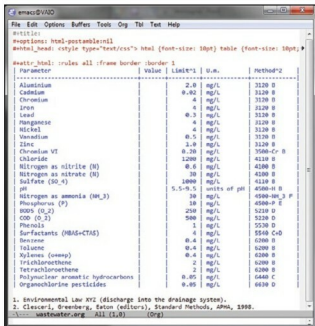
GNU Emacs, among other things, has a very powerful org-mode. Building and formatting tables in this mode is very fast (<http://orgmode.org/worg/org-tutorials/tables.html>). The result is a text file that can be read with a lot of other software. Here, the possibility of exporting a file from org to html is a very useful feature. The *sample notebook* (Figure 3) is an org-file containing the parameters to be measured on the sample and some other information about it. The *analysis report* (Figure 4) is an enlarged version of the notebook with other information also (for example, the customer's name and report comments, if any). The notebook and the report's 'htmlized' file paths are both stored in the database (the word 'htmlized' is a bit unusual, and is taken from <http://orgmode.org/manual/Publishing-action.html>). The files on the hard disk can be organised into two directories and some sub-directories: notebooks (or templates) and reports, each with sub-directories (for example: generic, customer A, customer B, customer ...).

Print with PHP

A printable version of the notebook and of the report can be built using a PHP script. This script contains a query to extract the information to be put on the document and includes the html file corresponding to the notebook or the report. The two tables shown in Figure 5 have been a little shortened to have a document of one page (two pages would be too long to be presented here) and the numeric values are for explanation purposes only. The methods are taken from Reference 2 and the *report notes* from Reference 3, given at the end of the article. If an error is recognised when the report has already been sent to the customer, to trace the two different versions of the printed report, the date and time of printing are at the bottom of the page. In this case, it's appropriate to also take some notes in the *field notes* and/or in the *field report notes*.

Due to the high flexibility of this system, I can write all the PHP scripts that I need and my www directory on Uniform Server can contain a lot of PHP files (Figure 6). The same holds for the SQL queries to be used directly in HeidiSQL. Because the database is used mostly for reading information from it, the PHP files are very useful to view the tables and to extract information from them. HeidiSQL can be used mainly as writing software. Obviously, as an alternative, more complex PHP scripts (or other languages/technologies) can be written to edit and add/delete records. For example, the information can be summarised by 'list' or by 'form'. A 'list' is a table containing all the records, in which each record is listed in one row and the columns are composed of some particular fields only. A 'form' is a table containing all the fields for each record (something like a key-value table). This is explained by the following code; in the 'list', only seven fields are used while the 'form' has 21 (all) fields:

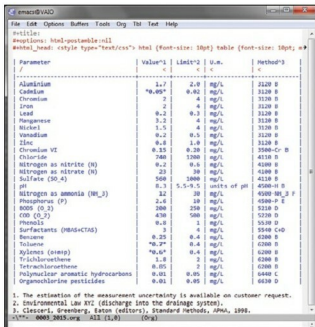
```
// Inventory "list"
```



Parameter	Value1	Limit1	U.n.	Method2
Aluminium	2.0	ng/L		3120 B
Cadmium	0.92	ng/L		3120 B
Chromium	4	ng/L		3120 B
Iron	4	ng/L		3120 B
Lead	0.2	ng/L		3120 B
Manganese	4	ng/L		3120 B
Nickel	0.2	ng/L		3120 B
Vanadium	0.2	ng/L		3120 B
Zinc	1.0	ng/L		3120 B
Chromium VI	0.20	ng/L		3500-Cr B
Chloride	1200	ng/L		4110 B
Nitrogen as nitrite (N)	0.5	ng/L		4100 B
Nitrogen as nitrate (N)	30	ng/L		4100 B
Sulfate (SO ₄)	500	ng/L		5110 B
pH	5.5-9.5	units of pH		4500-B E
Nitrogen as ammonia (NH ₃)	10	ng/L		4500-B E
Phosphorus (P)	10	ng/L		4500-B E
RODS (O ₂)	250	ng/L		5110 B
COD (O ₂)	500	ng/L		5110 B
Phenols	1	ng/L		5510 B
Surfactants (MBAS-CTAS)	3	ng/L		5510 C+D
Benzene	0.25	ng/L		6200 B
Toluene	0.4	ng/L		6200 B
Xylenes (oemp)	0.4	ng/L		6200 B
Trichloroethene	1.0	ng/L		6200 B
Tetrachloroethene	0.05	ng/L		6200 B
Polymuclear aromatic hydrocarbons	0.01	ng/L		6400 C
Organochlorine pesticides	0.05	ng/L		6410 B

1. Environmental law XYZ (discharge into the drainage system).
 2. Clesceri, Greenberg, Eaton (editors), Standard Methods, APHA, 1998.
 3. www.water.org All (1,0) (Org)

Figure 3: Example of a sample notebook



Parameter	Value1	Limit1	U.n.	Method3
Aluminium	2.0	ng/L		3120 B
Cadmium	0.92	ng/L		3120 B
Chromium	4	ng/L		3120 B
Iron	4	ng/L		3120 B
Lead	0.2	ng/L		3120 B
Manganese	4	ng/L		3120 B
Nickel	0.2	ng/L		3120 B
Vanadium	0.2	ng/L		3120 B
Zinc	1.0	ng/L		3120 B
Chromium VI	0.20	ng/L		3500-Cr B
Chloride	1200	ng/L		4110 B
Nitrogen as nitrite (N)	0.5	ng/L		4100 B
Nitrogen as nitrate (N)	30	ng/L		4100 B
Sulfate (SO ₄)	500	ng/L		5110 B
pH	5.5-9.5	units of pH		4500-B E
Nitrogen as ammonia (NH ₃)	10	ng/L		4500-B E
Phosphorus (P)	10	ng/L		4500-B E
RODS (O ₂)	250	ng/L		5110 B
COD (O ₂)	500	ng/L		5110 B
Phenols	1	ng/L		5510 B
Surfactants (MBAS-CTAS)	3	ng/L		5510 C+D
Benzene	0.25	ng/L		6200 B
Toluene	0.4	ng/L		6200 B
Xylenes (oemp)	0.4	ng/L		6200 B
Trichloroethene	1.0	ng/L		6200 B
Tetrachloroethene	0.05	ng/L		6200 B
Polymuclear aromatic hydrocarbons	0.01	ng/L		6400 C
Organochlorine pesticides	0.05	ng/L		6410 B

1. The estimation of the measurement uncertainty is available on customer request.
 2. Environmental law XYZ (discharge into the drainage system).
 3. Clesceri, Greenberg, Eaton (editors), Standard Methods, APHA, 1998.
 3. www.water.org All (1,0) (Org)

Figure 4: Example of an analysis report

```
<?php
$con=mysql_connect("localhost","root","query");
mysql_select_db("lims",$con);
$query=mysql_query("your-query-here");
mysql_close($con);
$num_rows=mysql_num_rows($query);
echo"Total: $num_rows records";
while($row=mysql_fetch_assoc($query)) {
    echo"<tr>";
    echo"<td>".$row['id'].</td>";
```